# Supplement: How Much Can Conjunction Message Histories Predict Beyond Persistence?

Arjun Vijay Prakash

CDM rows → cutoff-safe histories → event-disjoint split
→ candidate models → validation selection
→ floor hurdle + split conformal → frozen test / official test

Fig. 1. From cutoff-safe CDMs to a frozen test. Validation selects the floor hurdle; official-test scores are reports only.

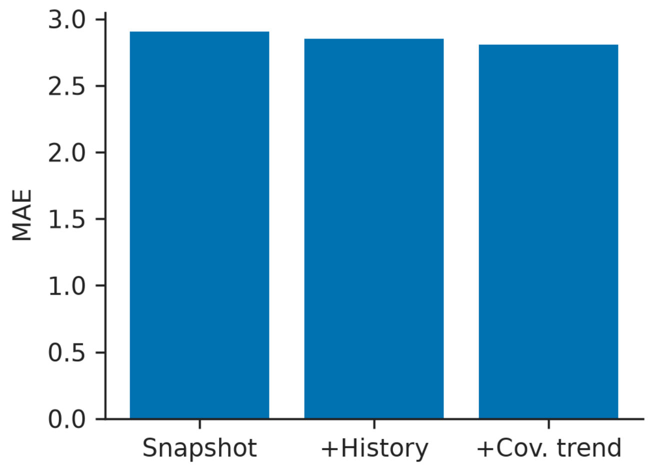


Fig. 2. Unguarded XGBoost MAE: snapshot, plus history, plus covariance trends.

This supplement holds overflow tables and figures for the IEEE conference paper. Confirmatory claims remain in the main text. Code: https://github.com/ArjunCodess/PRISM.

## I. Hypothesis checklist

- H1: matched-cohort $\Delta$MAE $4.49 \to 0.03$ — supported, with the 12 h CI covering zero.
- H2: floor-excluded MAE worse than persistence at the primary split, and non-floor $\Delta$MAE negative at every matched horizon — supported.
- H3: ESA $L$, LOO 66/66, official test — supported under this protocol.
- H4a: $\log\det$ vs movement $\rho = 0.399$ — univariate yes; partial coefficient reverses after controlling for risk.
- H4b: gap vs floor AUC 0.819 — supported.
- H4c: gap vs movement $\rho = -0.768$ — supported (negative).

## II. Protocol

Fig. 1 is the leakage-safe pipeline. All model and policy choices were frozen before official-test evaluation.

## III. Feature families

The shipped feature table has 234 numeric columns: 77 snapshot (CDM fields plus derived geometry and object-type dummies), 68 history summaries, and 88 covariance trends. History stems are risk, miss distance, relative speed, max-risk estimate, and observation counts. Covariance stems are RTN sigmas and log position-covariance determinants. All rows have $\mathrm{time\_to\_tca}$ at or above the cutoff. `dilution_gap` is excluded from the shipped booster. The full name list is `ml/artifacts/feature_schema.json`.

## IV. Ablation

Snapshot MAE 2.904; after history 2.851; after covariance trends 2.808. History adds a small increment relative to the snapshot itself.

## V. Floor-classifier thresholds

Thresholds 0.05 / 0.10 / 0.15 / 0.20 / 0.30 / 0.50 were scored on the frozen hurdle probabilities without retuning. The validation-selected operating point remains 0.15 (TP 1319, FP 174, FN 33, TN 133). False positives have median later $y = -14.3$ and mean snapshot risk $-14.9$.

## VI. Five-seed redraws

Seeds 42–46 retrain unguarded and residual XGBoost on grouped event splits. Unguarded MAE advantage $2.34 \pm 0.04$; floor-excluded MAE $7.78 \pm 0.21$; persistence floor-excluded $4.32 \pm 0.23$. Seed 42 remains the reported row.

## VII. Unmatched horizon table

The overlapping-set table in earlier drafts mixed cohort change with horizon: advantages 4.534 / 2.272 / 0.524 / 0.060 at 72 / 48 / 24 / 12 h. The matched-cohort table in the main paper is the H1 result.

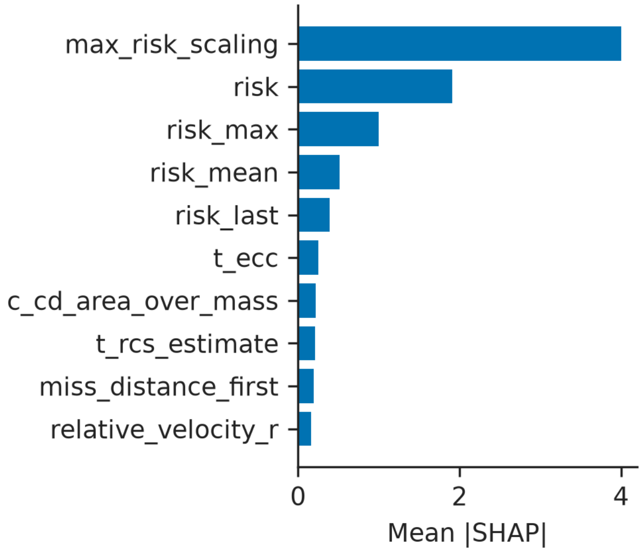


Fig. 3. Mean |SHAP| of the residual regressor on 80 test rows. Risk-trajectory fields dominate; this is not a physical cause.

## VIII. Gap quartiles

Train-quartile edges of `max_risk_estimate−risk` on the frozen test: Q1 floor rate 0.556 and mean $|y - \mathrm{risk}| = 13.03$; Q3–Q4 floor rates above 0.95 and mean movement below 0.75.

## IX. Residual SHAP

## X. Reproducibility checklist

- Code: https://github.com/ArjunCodess/PRISM
- Dataset: ESA Kelvins training archive and Zenodo 4463683
- Split seed 42; IDs in `ml/artifacts/split_manifest.json`
- XGBoost: 250 trees, depth 4, learning rate 0.05, subsample 0.85, colsample 0.8, min-child-weight 6, $\ell_1 = 0.1$, $\ell_2 = 2.0$, no early stopping, native NaN
- Floor classifier: 180 trees, depth 3, threshold 0.15, no persist guard
- Figures: `python ml/src/plots.py`
- Official test frozen before look; no retune